

Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

Prepared for APL Logistics (KWE Group)

Unified Mentor — Data Analytics Internship Project

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Dataset: 180,519 raw order records · 172,765 analyzed after cleaning

Executive Abstract

APL Logistics manages high-volume, multi-region shipments, yet lacked a systematic view of on-time performance and the structural drivers behind late deliveries. This analysis examines 172,765 cleaned order records spanning five global markets, four shipping modes, and 23 order regions to quantify delivery performance and diagnose late_delivery_risk.

The central finding is that delay risk in this operation is overwhelmingly structural, not geographic. On-time delivery stands at only 18.6% of orders, with 57.3% classified as delayed. Shipping mode explains almost all of the variation: First Class shipments are late 100% of the time because they are scheduled for 1 day but consistently take 2 days to fulfill — a fixed, systemic SLA mismatch rather than a random operational failure. By contrast, delay risk is nearly flat across regions, markets, and product categories (a 55%–70% band), indicating that fixing the shipping-mode/schedule mismatch would have far greater impact than any single regional intervention.

1. Background and Context

In large-scale global logistics operations, delivery delays lead to Service Level Agreement (SLA) violations and financial penalties. A mismatch between planned and actual shipping time increases operating cost, inefficient shipping modes reduce customer satisfaction, and regional bottlenecks are frequently detected only after they have already affected customers. APL Logistics, operating under the KWE Group, handles high-volume, multi-region shipments across five markets (LATAM, Europe, Pacific Asia, USCA, and Africa), making delivery-performance monitoring a critical part of operational excellence.

This project was undertaken as part of the Unified Mentor internship program, with the goal of providing operational transparency into delivery timelines and delay risk ahead of any future predictive or optimization modeling work.

2. Problem Statement

Despite having detailed order and shipping data, the organization lacked:

- A clear, quantified measurement of on-time versus delayed deliveries
- An understanding of why shipments are delayed
- Visibility into which regions, markets, and shipping modes carry the highest risk
- Diagnostics that explain the drivers behind late_delivery_risk

As a result, logistics teams have historically operated reactively — responding to delays after they occur — rather than preventively.

3. Data and Methodology

3.1 Dataset

The dataset contains 180,519 raw order-line records across 40 fields, covering order timing, customer attributes, product/category detail, shipping mode, and financials (sales, profit, discount). After removing cancelled orders and invalid shipping-duration records, 172,765 rows (95.7% of the raw data) were retained for analysis.

3.2 Data Cleaning & Validation

- Removed orders with status "Shipping canceled" or "CANCELED" (7,754 rows, 4.3%) — these have no genuine delivery outcome to measure.
- Validated that shipping duration fields (real and scheduled) fall within a sane 0–60 day range.

- Standardized text fields (region, market, shipping mode, segment) by trimming whitespace for consistent grouping.
- Filled minor missing values in non-analytical fields (8 missing last names, 3 missing zip codes) without dropping rows.

3.3 Delivery Gap Calculation

The core diagnostic metric is the Delivery Delay Gap, computed as:

$$\text{Delay Gap} = \text{Days for shipping (real)} - \text{Days for shipment (scheduled)}$$

Each order was then classified as:

- On-time — Delay Gap = 0
- Delayed — Delay Gap > 0 (shipped later than scheduled)
- Early — Delay Gap < 0 (shipped ahead of schedule)

4. Overall Delivery Performance

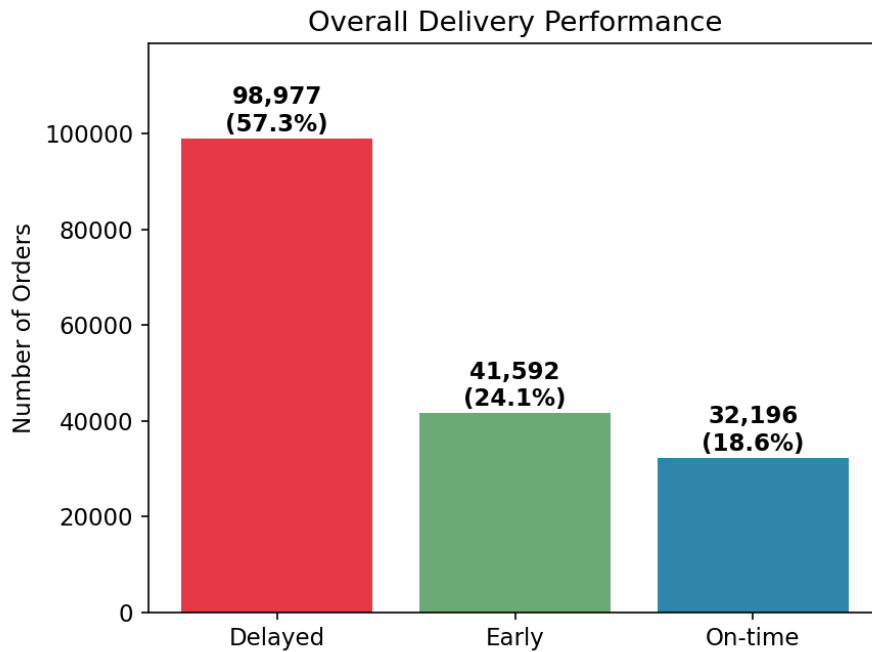


Figure 1. Overall delivery outcome distribution across 172,765 orders.

KPI	Value
On-Time Delivery Rate	18.64%
Delayed Delivery Rate	57.29%
Early Delivery Rate	24.07%
Average Delivery Delay Gap	+0.57 days
Late Delivery Risk Ratio	57.29%

Fewer than one in five orders is delivered exactly on schedule. Over half of all orders (57.3%) are delivered later than promised, while roughly a quarter arrive early. Combined with the shipping-mode breakdown below, this points to a baseline scheduling problem rather than isolated operational failures.

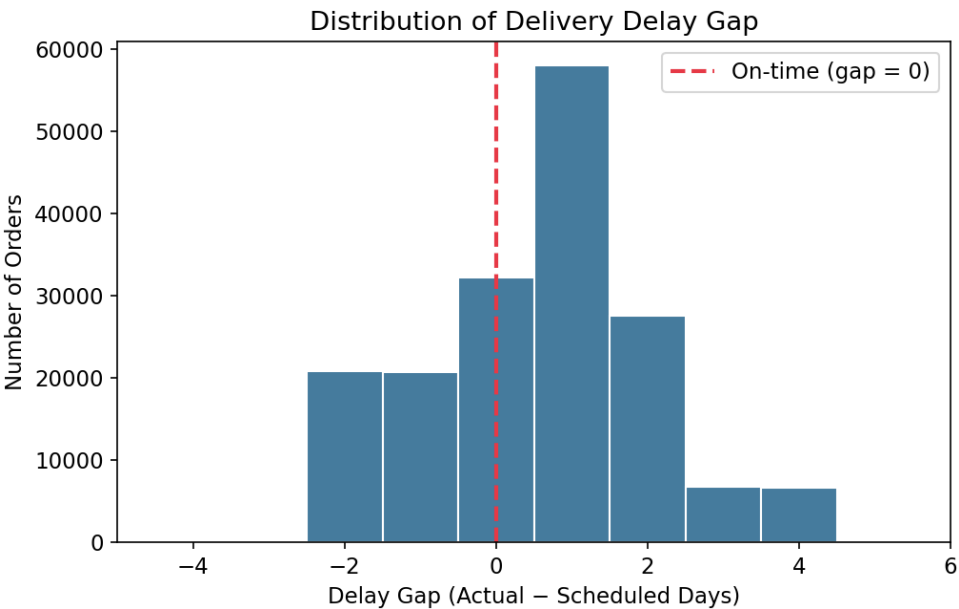


Figure 2. Distribution of the delay gap. The tallest bars sit to the right of zero, confirming that late deliveries dominate.

5. Shipping Mode Efficiency Analysis

Shipping mode is the single strongest driver of delay risk identified in this analysis — far stronger than geography, product category, or customer segment.



Figure 3. Late delivery risk by shipping mode.

Shipping Mode	Orders	Avg Delay Gap (days)	Late Risk %	On-Time %
First Class	26,513	1.00	100.00%	0.00%
Second Class	33,806	1.99	79.83%	20.17%
Same Day	9,293	0.48	47.93%	52.07%
Standard Class	103,153	-0.01	39.77%	19.91%

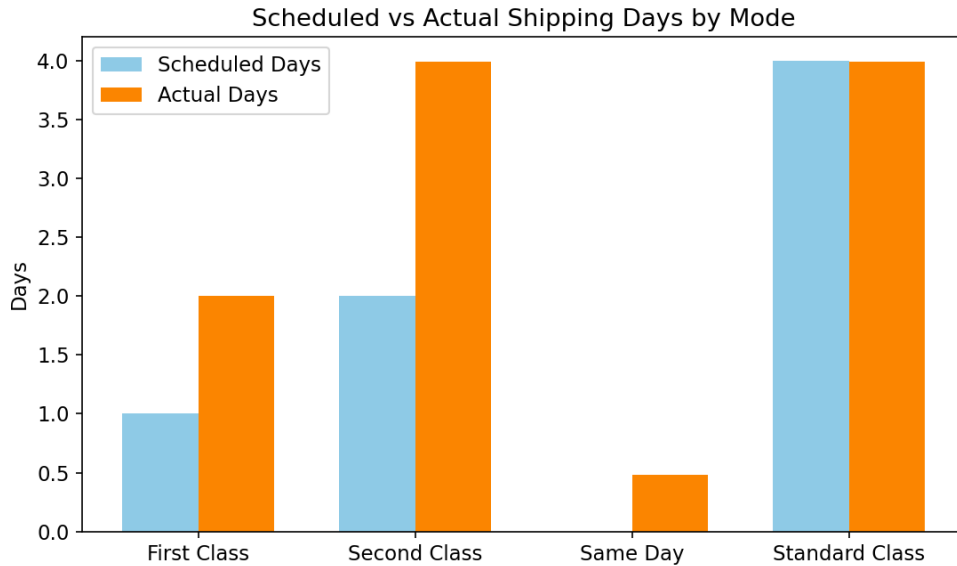


Figure 4. Scheduled vs. actual shipping days by mode. First Class is always scheduled for 1 day but always actually takes 2.

The First Class finding is the most actionable insight in this report: every single First Class order (100%) is late, not due to operational inconsistency, but because the promised 1-day window is structurally unachievable — actual fulfillment for this mode is uniformly 2 days. This is a fixed SLA design flaw, not a variable operational risk, and it can be corrected immediately by either re-promising First Class as a 2-day service or re-engineering fulfillment to genuinely support 1-day delivery.

Second Class also carries high risk (79.8%) because its scheduled window (2 days) is shorter than its actual fulfillment spread, which extends fairly evenly from 2 to 6 days. Standard Class, with the most generous schedule (4 days), is the best-performing mode on a risk basis (39.8% late) even though it carries the highest order volume.

6. Regional & Market Diagnostics

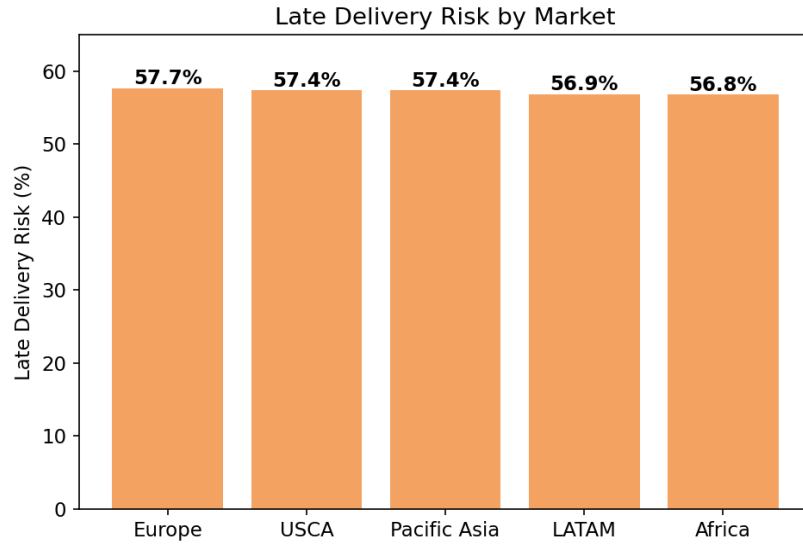


Figure 5. Late delivery risk by market.

Market	Orders	Avg Delay Gap	Late Risk %
Europe	48,090	0.57	57.69%
USCA	24,627	0.58	57.41%
Pacific Asia	39,585	0.57	57.38%
LATAM	49,309	0.55	56.87%
Africa	11,154	0.56	56.84%



Figure 6. Late delivery risk by order region (top 15).

Late delivery risk is remarkably uniform across markets, varying by less than one percentage point between the highest (Europe, 57.7%) and lowest (Africa, 56.8%). At the more granular Order Region level, risk ranges from roughly 55% to 60% across the great majority of regions, with only a handful of small-volume outliers (e.g., Central Africa at 60.2%) at the edges. At the country level, even the highest-risk countries with meaningful volume — Ecuador, Mozambique, and Chile — sit in the 63–71% range, still within a similar overall band.

This is an important negative finding for operations planning: geography is not a meaningful lever for reducing delay risk in this dataset. Because the shipping-mode effect (Section 5) dwarfs any regional effect, corrective investment aimed at specific regions is unlikely to move the needle nearly as much as fixing the mode-level scheduling mismatch.

7. Customer Segment Impact Analysis

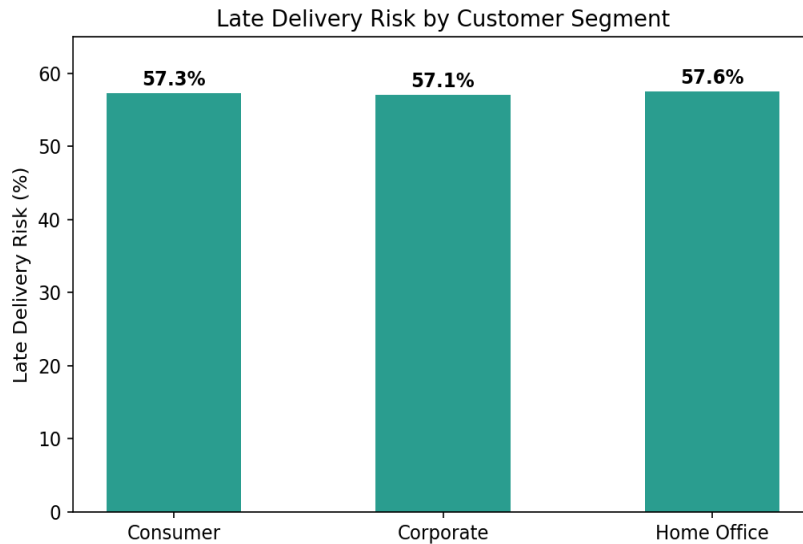


Figure 7. Late delivery risk by customer segment.

Segment	Orders	Avg Delay Gap	Late Risk %	Avg Sales (\$)	Avg Profit (\$)
Consumer	89,420	0.57	57.31%	204.25	22,17
Corporate	52,528	0.56	57.08%	203.92	22,04
Home Office	30,817	0.58	57.59%	202.44	21,60

Customer segment shows essentially no differentiation in delay risk (56.9%–57.6% across all three segments) or in average order value. This means higher-value Corporate and Home Office accounts are exposed to the same SLA risk as standard Consumer orders — there is no evidence that premium customers are being systematically protected from delays, which represents a latent account-risk exposure worth flagging to account management.

8. Financial Impact of Delivery Outcomes

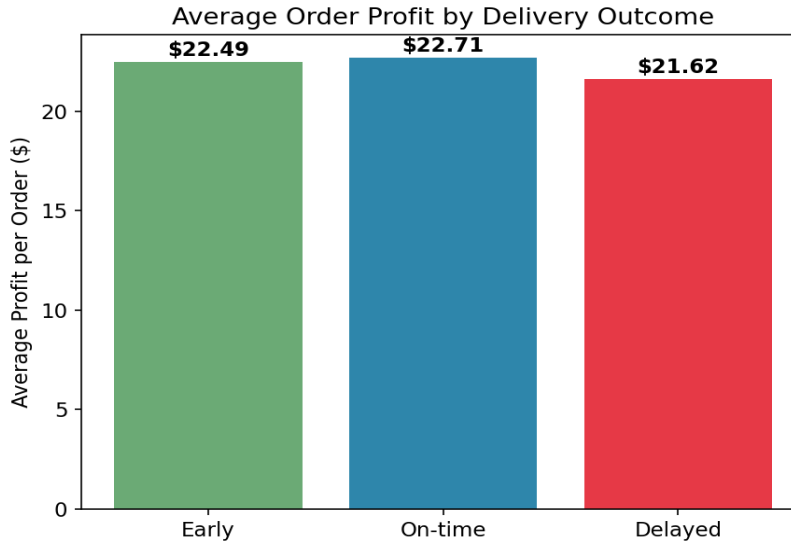


Figure 8. Average profit per order by delivery outcome.

Delayed orders earn slightly less average profit per order (\$21.62) than on-time (\$22.71) or early (\$22.49) orders — roughly a 4.8% profit gap versus on-time orders. Because delayed orders are also by far the largest group by volume, they account for the majority of total profit in absolute terms (\$2.14M of the total observed), but on a per-order basis they are the least profitable outcome, consistent with the operational cost of expediting, re-routing, or servicing SLA complaints for late shipments.

9. Key Findings Summary

- Only 18.6% of orders are delivered exactly on schedule; 57.3% are late.
- Shipping mode is the dominant driver of delay risk: First Class is 100% late by design (1-day promise, 2-day actual fulfillment).
- Standard Class, despite having the largest volume, has the lowest late-risk rate (39.8%) due to its more realistic 4-day schedule.
- Region, market, and country have only marginal influence on delay risk (a narrow 55%–70% band) compared to shipping mode.
- Customer segment (Consumer / Corporate / Home Office) shows no meaningful difference in delay exposure — premium accounts are not currently protected.
- Delayed orders are ~4.8% less profitable per order than on-time orders.

10. Recommendations

- Re-align First Class SLA promises to 2 days, or invest in fulfillment capacity so 1-day delivery becomes achievable — this single change would eliminate the highest-risk mode's 100% late rate.
- Review Second Class scheduling; either extend the promised window or tighten fulfillment consistency to reduce its 79.8% late rate.
- Do not prioritize region-specific delay interventions over mode-level fixes — the data shows geography is not a major differentiator.
- Introduce segment-aware SLA monitoring so Corporate and Home Office accounts, which carry higher account value, can be flagged for proactive communication when a delay risk is detected.

- Use the accompanying Streamlit dashboard as a continuous monitoring layer, and treat this analysis as the diagnostic baseline before building a predictive `late_delivery_risk` model.

11. Conclusion

This project establishes a diagnostic intelligence layer for APL Logistics' supply chain operations. The evidence points to a clear, addressable root cause for the majority of delivery delays: shipping-mode SLA promises that do not match actual fulfillment capability, rather than regional bottlenecks or customer-segment effects. Correcting the mode-level scheduling mismatch — particularly for First Class and Second Class — offers the highest-leverage path to improving on-time performance. This analysis forms the critical first step before deploying predictive or optimization-based supply chain models.